Convolutional VAEs for Chest X-ray Generation

Vincent Chen
ECE 285 Homework 1, UC San Diego

Abstract

We train convolutional VAEs on pediatric chest X-rays and score samples with FID and Inception Score. A sweep over the KL weight $\beta \in \{1, 0.1, 0.01\}$ shows that reconstructions improve as β falls (FID 221 to 134), while prior samples improve only at β=0.01 (219, 220, 185). Sampling from a Gaussian mixture fitted to the training latents helps once β is small and closes about half of the gap to reconstructions; the best model (β=0.01, mixture sampling) reaches FID 158. An ablation shows that rotation/translation augmentation hurts (FID 214), which we attribute to dark corner artifacts from the black fill of rotated images.

1 Introduction

We implement a VAE Kingma and Welling [2013] in PyTorch and train it on the Chest X-Ray Images (Pneumonia) dataset Kermany et al. [2018]. The closest prior work we found Chandra et al. [2025] trains a similar 64×64 VAE on this data with a very small KL weight (β=0.005) and judges samples by eye. We add FID Heusel et al. [2017] and Inception Score Salimans et al. [2016], and measure how β and the sampling distribution change them.

2 Method

Model. Encoder: four stride-2 4×4 convolutions (64, 128, 256, 512 channels) with BatchNorm and LeakyReLU(0.2), then two linear heads for the mean and log-variance (clamped to $[-10, 10]$) of a 128-d Gaussian posterior. The decoder mirrors it with transposed convolutions and a sigmoid output (8.7M parameters).

Loss and sampling. $\mathrm{BCE}(x, \hat{x}) + \beta\, D_{\mathrm{KL}}(q(z|x) \,\|\, \mathcal{N}(0, I))$, BCE summed over pixels; β grows linearly from 0 over the first 10 epochs to avoid posterior collapse. "Prior" samples decode $z \sim \mathcal{N}(0, I)$. Since small β leaves the aggregate posterior far from $\mathcal{N}(0, I)$, "GMM" samples instead decode z from a 10-component Gaussian mixture fitted to the training latent means Ghosh et al. [2019].

3 Setup

Data. The 5,216 training images are split 90/10 (seeded) into 4,694 train and 522 validation images; the 624 test images are the FID reference. Images are grayscale, resized and center-cropped to 64×64, scaled to $[0, 1]$; no augmentation is used (ablation below).

Training and evaluation. Adam, lr 10^{-3} (halved on validation plateaus), batch 128, 100 epochs, seed 0, on an Apple M1 Pro GPU (≈8 s/epoch); the lowest-validation-loss checkpoint is evaluated. FID and IS use 2,000 generated images, PyTorch-Ignite[1], and ImageNet InceptionV3 Szegedy et al. [2016] features (grayscale replicated to 3 channels, resized to 299×299); IS is the mean±std over 10 splits. "Recon." is the FID of the 624 test-set reconstructions, a reference point. FID is biased upward for small sample sizes, so "GMM@624" scores only 624 GMM samples, to compare like with like against Recon.

[1] `https://github.com/pytorch/ignite`

Table 1: FID (lower is better) against the 624 test images. Prior and GMM use 2,000 generated images; GMM@624 and Recon. use 624. Inception Score (higher is better) is for prior samples.

β	FID prior	FID GMM	FID GMM@624	FID recon.	IS prior
1	219.2	220.8	217.8	220.6	1.91 ± 0.04
0.1	220.3	206.5	202.6	201.4	1.84 ± 0.06
0.01	184.8	157.9	159.3	134.4	1.70 ± 0.04

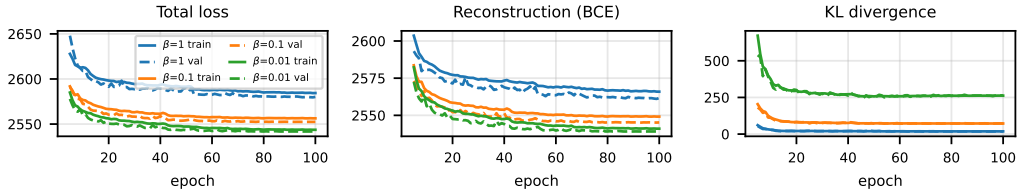


Figure 1: Training (solid) and validation (dashed) losses; epochs 1–4 (KL warm-up) omitted.

4 Results

Training. All runs converge smoothly (Fig. 1); smaller β lowers validation BCE (2562, 2545, 2539 for $\beta = 1, 0.1, 0.01$) but raises KL (18, 72, 263 nats).

Quantitative. In Table 1, reconstruction FID improves as β falls (221, 201, 134); prior FID is flat (219, 220) until β=0.01 (185). GMM sampling helps only for $\beta \leq 0.1$ (220→207, 185→158): at β=1 the aggregate posterior is already close to $\mathcal{N}(0, I)$ (KL 18 nats), so GMM≈prior. At β=0.01 it closes about half of the 50-point gap to reconstructions (134). At matched size (GMM@624: 218, 203, 159), GMM matches reconstructions at β=1 and 0.1 and trails only at β=0.01 (159 vs. 134). IS of prior samples falls (1.91, 1.84, 1.70) even as FID holds or improves; since ImageNet classes separate these images poorly, we treat it as uninformative. **Augmentation ablation** (β=0.01, rotation $\pm 5°$, translation 5%): FID prior/GMM/recon. rises from 185/158/134 to 323/214/206 although validation loss (BCE + βKL) is equal (2542), which alone would not reveal the degradation.

Qualitative. Prior samples of the β=0.01 model (Fig. 2, left) show chest outlines but are noisier than GMM samples (middle), which have clearer rib cages and lung fields. With augmentation, prior samples had dark borders (mean brightness of the outer 4 pixels: 0.34 vs. 0.54 for test images; 0.52 without augmentation), consistent with the black fill of rotated training images; the augmented model's training loss was also 118 nats below validation (2 without). Reconstructions lose rib detail; a PCA of test latents shows heavy NORMAL/PNEUMONIA overlap. Limitations: blurry output, one seed, untuned GMM size, no downstream validation.

AI usage statement

Claude Code (Anthropic) wrote the code and tests, ran training and evaluation on the author's laptop, checked the references, and drafted this report; a separate Claude review of the code and report led to two fixes. The author found the related paper, chose the setup, approved the design, plan and downloads, and reviewed the results.

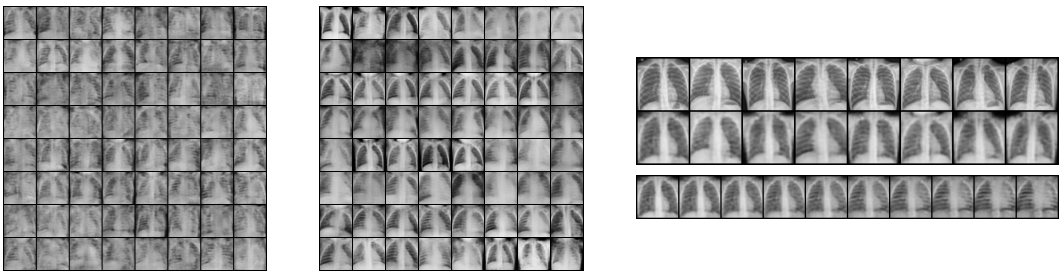


Figure 2: β=0.01 model. Left: prior samples. Middle: GMM samples. Right: test images over their reconstructions (top); latent interpolation from a normal to a pneumonia image (bottom).

References

Ellen Chandra, Natania Anita, Olivia Tiffany, Nikita Ananda Putri Masaling, and Andry Chowanda. Synthetic chest X-ray generation using variational autoencoders for pneumonia disease. In *International Conference on Computer Engineering, Network, and Intelligent Multimedia (CENIM)*, 2025. doi: 10.1109/CENIM67940.2025.11326136.

Partha Ghosh, Mehdi S. M. Sajjadi, Antonio Vergari, Michael Black, and Bernhard Schölkopf. From variational to deterministic autoencoders. *arXiv preprint arXiv:1903.12436*, 2019.

Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems*, 2017.

Daniel S. Kermany, Michael Goldbaum, Wenjia Cai, et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131, 2018.

Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.

Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training GANs. In *Advances in Neural Information Processing Systems*, 2016.

Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2016.